

Cultural traits can evolve. Here we use a model of vertical cultural transmission developed by Cavalli-Sforza & Feldman 1981, where a dichotomous cultural trait (with types b and B) is passed from parents to offspring. The cultural type of the offspring is determined by a set of transmission parameters, g_i , as shown in the following table.

Mating type		Fraction of	
parent 1	parent 2	B offspring	b offspring
b	b	0	1
b	B	g_1	$1 - g_1$
B	b	g_2	$1 - g_2$
B	B	1	0

Let $x(t)$ be the frequency of the cultural innovation, B , in generation t . Assuming random mating, the frequency of the cultural innovation in the next generation is then

$$x(t+1) = x(t)(1-x(t))g_1 + x(t)(1-x(t))g_2 + x(t)^2.$$

1. [6/20 points] Show how to find the two equilibria, $\hat{x} = 0, 1$.

We set $x(t+1) = x(t) = \hat{x}$ and solve for $\hat{x}$,

$$\hat{x} = \hat{x}(1-\hat{x})g_1 + \hat{x}(1-\hat{x})g_2 + \hat{x}^2.$$

We see that $\hat{x} = 0$ satisfies this equation, and so is an equilibrium. If $\hat{x} \neq 0$ we can divide both sides by $\hat{x}$,

$$\begin{aligned} 1 &= (1-\hat{x})g_1 + (1-\hat{x})g_2 + \hat{x} \\ 1-\hat{x} &= (1-\hat{x})g_1 + (1-\hat{x})g_2. \end{aligned}$$

This equation is satisfied by $\hat{x} = 1$, and so is our second equilibrium.

2. [4/20 points] Given g_1 and g_2 are fractions, what are the minimum and maximum values $g_1 + g_2$ can take?

A fraction is bounded between 0 and 1, so $0 \leq g_i \leq 1$ for $i = 1, 2$. This means that the minimum value $g_1 + g_2$ can take is 0 and the maximum value it can take is 2.

3. [8/20 points] Show how to determine that $\hat{x} = 0$ is only stable when $g_1 + g_2 < 1$ and $\hat{x} = 1$ is only stable when $g_1 + g_2 > 1$.

To determine stability we need to know the slope of the recursion,

$$\frac{dx(t+1)}{dx(t)} = (1 - 2x(t))g_1 + (1 - 2x(t))g_2 + 2x(t).$$

We then evaluate it at the equilibria. First $x(t) = \hat{x} = 0$,

$$\left. \frac{dx(t+1)}{dx(t)} \right|_{x(t)=0} = g_1 + g_2.$$

Stability requires that this is less than 1 in absolute value, $|g_1 + g_2| < 1$, equivalently $-1 < g_1 + g_2 < 1$. From the previous question we know that $0 \leq g_1 + g_2$, so the lower inequality is guaranteed and the only condition we need to satisfy is $g_1 + g_2 < 1$.

We do the same for the other equilibrium, $x(t) = \hat{x} = 1$. The of slope of the recursion at this point is

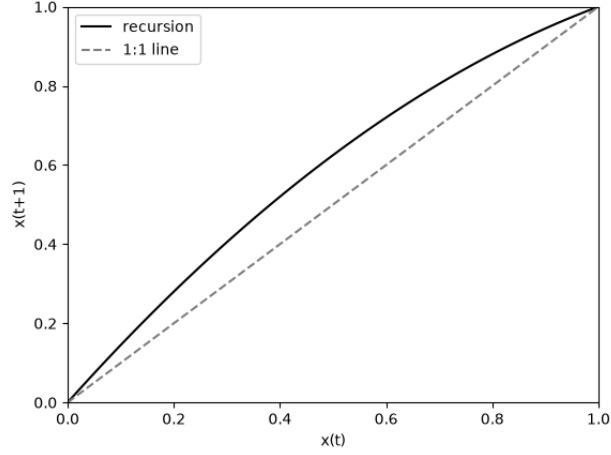
$$\left. \frac{dx(t+1)}{dx(t)} \right|_{x(t)=1} = -g_1 - g_2 + 2,$$

which means we need

$$\begin{aligned} |2 - (g_1 + g_2)| &< 1 \\ -1 &< 2 - (g_1 + g_2) < 1 \\ -3 &< -(g_1 + g_2) < -1 \\ 1 &< g_1 + g_2 < 3 \end{aligned}$$

for stability. From the previous question we know $g_1 + g_2 \leq 2$ so we only need to check that we satisfy the lower bound, $1 < g_1 + g_2$.

4. [2/20 points] Which equilibrium is stable in the plot below?



We see that the recursion is above the 1:1 line (except at the equilibria, $\hat{x} = 0, 1$, where the two lines cross). In other words, when we are not at equilibrium $x(t+1)$ is greater than $x(t)$, meaning x is increasing. This implies that the upper equilibrium, $\hat{x} = 1$, is stable for the parameter values used to make this plot. Another way to determine that $\hat{x} = 1$ is stable in this plot is to notice that the slope of the recursion through $x(t) = 1$ is between -1 and 1 (while the slope through $x(t) = 0$ is greater than 1, implying instability).